Research in International Business and Finance 73 (2025) 102575

Contents lists available at ScienceDirect

Research in International Business and Finance

journal homepage: www.elsevier.com/locate/ribaf

A hybrid deep learning model for cryptocurrency returns forecasting: Comparison of the performance of financial markets and impact of external variables

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ARTICLE INFO

Keywords:
Forecasting
Financial markets
Blockchain information
Twitter economic uncertainty
CO2 emissions

ABSTRACT

This study introduces a finetuned hybrid forecasting model combining both Discrete Wavelet Transform (DWT) and Long Short-Term Memory network (LSTM) to predict dirty and clean cryptocurrency returns (Bitcoin and Ripple). The findings show that the proposed DWT-LSTM model outperforms a large set of benchmark models in terms of forecasting accuracy. We investigate a broader set of predictors involving financial markets (other cryptocurrencies and commodities) and external variables (blockchain information, Twitter economic uncertainty, and CO2 emissions). Our findings underline the comparable performance of the considered predictors, with the Twitter Economic Uncertainty index being the best predictor of Bitcoin returns and S&P GSCI Energy being the best predictor of Ripple returns. We also highlight the superior performance of the trading strategies based on our forecasting results.

1. Introduction

Cryptocurrencies are characterized by high volatility, noisy price fluctuations, and price bubbles (Chaim and Laurini, 2018; Hafner, 2018; Katsiampa, 2017). Determining the factors that affect the monetary value of cryptocurrencies is essential for forecasting future prices and estimating expected returns (Li and Wang, 2017; Mai et al., 2018; Li et al., 2019; Guo et al., 2021; Lyócsa et al., 2020). Cryptocurrency returns could potentially be forecasted by considering several driving factors, such as, number and cost of transactions, trading volume, Google trends, sentiment in newspaper reports, environmental variables, other traditional markets and economic policy uncertainty (e.g., Kristoufek, 2013, 2015; Polasik et al., 2015; Balcilar et al., 2017; Hayes, 2017; Bouri et al., 2017a, 2017b, 2018a, 2018b; Demir et al., 2018; Ji et al., 2019; Cong and He, 2019; Ren and Lucey, 2022; Kubal and Kristoufek, 2022; Kraaijeveld and de Smedt, 2020; Xie et al., 2020, Clark et al., 2023). The escalating environmental concerns have led to a division within the cryptocurrency market, distinguishing between green (clean) and dirty categories based on the energy consumption involved in the mining process (Yousaf et al., 2023). Recent studies explore the relationship between cryptocurrencies and green financial assets (Arfaoui et al., 2023; Pham et al., 2022; Naeem and Karim, 2021). Ren and Lucey (2022) assert that dirty cryptocurrencies rely on PoW algorithms for consensus, leading to substantial energy consumption for both mining and transactions, while clean cryptocurrencies utilize various energy-efficient consensus algorithms.

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https://doi.org/10.1016/j.ribaf.2024.102575
Received 6 March 2024; Received in revised form 10 July 2024; Accepted 6 September 2024
Available online 13 September 2024
0275-5319/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

A literature review reveals a growing demand for the development of accurate forecasting models. Autoregressive models have been used to investigate the predictability of returns; however, these models have a limited capacity to create accurate forecasts because of the nonlinear and nonstationary nature of financial market movements (Hansen and Lunde, 2005). To solve nonlinear challenges, Machine Learning (ML) and Deep Learning (DL) algorithms[1] have attracted the interest of academics and practitioners owing to their ability to capture complex nonlinear relationships in input data and predict financial market movements without relying on traditional assumptions (Atsalakis et al., 2019). However, a review of previous studies employing machine learning methods reveals that most used manual tuning and fixed parameters which presents significant drawbacks including the need for specialized expertise, the potential for human bias, and a lack of flexibility.

Therefore, this study aims to address some of the limitations identified in the existing literature concerning model selection and random parameter tunning in cryptocurrency return prediction by proposing a finetuned hybrid deep learning model, Discrete Wavelet Transformation-Long Short-Term Memory (DWT-LSTM). This model is optimized by sampling random combinations of hyper-parameters to identify the optimal set that yields the best performance on the validation datasets. This model forecasts the returns of two types of cryptocurrencies (Bitcoin for dirty and Ripple for clean), considering financial markets (other cryptocurrencies and commodities) as predictors and additional variables related to blockchain information, Twitter economic uncertainty, and CO2 emissions.

The results demonstrate the performance of the finetuned DWT-LSTM model compared to other ML and DL models and highlight the comparable performance of the considered predictors, with the Twitter Economic Uncertainty index and S&P GSCI Energy being, respectively, the best predictors of Bitcoin and Ripple returns. This study also reports the implications of the forecasting results on Bitcoin and Ripple trading strategies, allowing for better risk management, lower maximum drawdowns, and remarkable Sharpe and Sortino ratios, which underlines the usefulness of our proposed model as a trading support system. The subsequent sections of this paper are organized as follows. Section 2 delves into the relevant literature. Section 3 outlines our proposed model. Section 4 provides details about the data used in this study and discusses the results. Finally, Section 5 draws the conclusion, offering policy implications and exploring potential avenues for future research.

2. Related literature

According to Chen et al. (2021), considering the various determinants of cryptocurrencies to predict their returns is necessary. Based on previous literature, these determinants can be classified into four main categories: The first category corresponds to economic factors i.e., commodities, foreign exchange rates, stock market indices and economic uncertainty (Qiu et al., 2021; Sun et al., 2020; Salisu et al.; 2023). The second category is related to technology factors, i.e., blockchain information such as block size, number of transactions per block, hash rate, and miner revenue (Guo et al., 2021; Gradojevic et al., 2021). Public attention (e.g., Tweets and Google trends) corresponds to the third category (Guo et al., 2021; Wang et al., 2022; Dias et al., 2022) whereas the last category involves environmental factors (Clark et al., 2023). Few studies have combined several factors to understand cryptocurrency market behavior (Zhong et al., 2023; Liu et al., 2023; Chen et al., 2021; Hajek et al., 2023). Table 1 summarizes the related literature investigating cryptocurrencies forecasting.

An increasing number of empirical studies have analyzed the predictors of cryptocurrencies by resorting to conventional statistical, econometric, ML, and DL models. For instance, Catania et al. (2019) rely on Dynamic Model Averaging to conduct a comparative analysis of several alternative econometric models for the forecasting of four cryptocurrencies (Bitcoin, Litecoin, Ripple, and Ethereum) from August 8, 2015, to December 28, 2017, using a set of predictors (stock indices, gold, and interest rates). They emphasize the notable enhancement in point forecasting through the utilization of combinations of univariate models. Additionally, there is a noteworthy improvement in density forecasting when employing a selection of multivariate models. Liu and Tsyvinski (2021) examine the predictors of Bitcoin, Ethereum, and Ripple and conclude that cryptocurrency returns are influenced by cryptocurrency network factors but not production factors, and that cryptocurrency-specific factors (momentum and investor attention) strongly predict their returns. Bouri et al. (2022) find that their proposed model, based on Standard Factor models and Regime-Switching Factor loadings, outperforms the benchmark random walk in forecasting the returns of Bitcoin, Ethereum, Ripple, Dogecoin, and Litecoin while considering their lagged returns and other cryptocurrencies as predictors over the period from November 2016 to September 2021. Using OLS regression, Clark et al. (2023) show that combined levels of CO2 emissions can effectively predict Bitcoin and Ethereum returns, underlining the growing influence of environmental considerations on the performance of these cryptocurrencies. On the other hand, various ML and DL models have recently been proposed to forecast cryptocurrency returns owing to their characteristics of easy feature engineering and good performance. Sun et al. (2020) resort to three machine learning models (SVM, RF, and LGBM) to forecast the cryptocurrency market returns using daily data of 42 cryptocurrencies combined with stock markets, crude oil and fiat currencies during the period from January 1, 2018, to June 30, 2018 and find that LGBM stands out to be the best forecasting model compared to the other models. Gradojevic et al. (2021) use RF to forecast one-day-ahead Bitcoin prices from May 11, 2015, to March 7, 2019. They find that Bitcoin was driven primarily by its trend and momentum from 2015 to 2017, whereas market sentiment factors such as fear, greed, Google search patterns, and trading volume played a dominant role in shaping its behavior in 2018 and 2019. Chen et al. (2021) use ANN and RF to reduce the subset of potential predictors by assessing the importance of economic as well as technological factors that are integrated into long LSTM to predict Bitcoin prices regardless of historical prices during the period from

[1] Random Forest (RF), Support Vector Machines (SVM), XGBoost (XGB), Light Gradient Boosting Machine (LGBM), Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM).

Table 1
Summary of the related literature.

Authors	Variables	Models	Results
Catania et al. (2019)	– Target: Bitcoin, Litecoin, Ripple, and Ethereum – Predictors: stock indices, gold, and interest rates	– Dynamic Model Averaging	– The combinations of all variables enhance the forecasting accuracy of cryptocurrencies.
Wang et al. (2022)	– Target: Bitcoin – Predictors: Google trends and tweets, as well as Bitcoin trading data	– LSTM	– Google trends and tweets, as well as Bitcoin trading data, improve the prediction accuracy of Bitcoin returns.
Bouri et al. (2022)	– Target: Bitcoin, Ethereum, Ripple, Dogecoin, Litecoin – Predictors: Bitcoin, Ethereum, Ripple, Dogecoin, Litecoin, Stellar, Ethereum Classic, Moreno, NEO, Waves, Dash, Decred, Zcash and NEM	– Factor models	– Factor models (specially regime-switching versions) outperforms the random-walk model.
Chen et al. (2021)	– Target: Bitcoin – Predictors: blockchain information, Google Trends, tweets, TSE100, DOW30, VIX, SSE, crude oil price, gold price, S&P500, and NASDAQ. CHF, EUR, GBP, JPY, CNY and US dollar.	– Artificial neural network – Random forest – LSTM	– Bitcoin prices are influenced by different economic and technology factors.
Cheng et al. (2024)	– Target: Bitcoin – Predictors: open, high, low, close, adjective close and volume of Bitcoin	– LSTM – SARIMA – Facebook prophet	– LTSM has a noticeable improvement compared to SARIMA and Facebook prophet.
Clark et al. (2023)	– Target: Bitcoin, Ethereum, Tether and TrueUSD – Predictors: global temperature anomalies and CO2 emissions, gold, WTI	– Predictive regression model	– CO2 emissions significantly predict the returns of Bitcoin and Ethereum.
Feng et al. (2024)	– Target: Bitcoin and Ethereum – Predictors: realized volatility, jump components of RV, realized positive semi-variance, realized negative semi-variance, global economic uncertainty index, the US economic uncertainty index, geopolitical risk index, consumer and producer price index in the US, crude oil ETF volatility index, gold ETF volatility index, effective federal funds rate, nominal broad U.S. dollar index, West Texas intermediate crude oil index, brent crude oil prices, Google trend, blockchain information.	– HAR-RV – Lasso – Ridge regression – Elastic net	– Regularization models based on dynamic tuning (Lasso-tune, Ridge-tune, Ela-tune) outperform the benchmark model HAR-RV in terms of performance. – Hash rate and the effective federal funds rate play significant forecasting role among the other variables.
Gradojevic et al. (2021)	– Target: Bitcoin – Predictors: technical indicators, fear & greed, Google search, and volume	– Random Forest	– Random Forest trained on technical indicators is better than the random walk and ARMAX. – Bitcoin is driven by its trend and momentum, fear, greed, Google searches, and volume.
Guo et al. (2021)	– Target: Bitcoin – Predictors: Bitcoin prices, blockchain information, Google trends	– Wavelet Transform (WT)-Casual Multi-Head Attention (CA)-Temporal Convolutional Network (TCN)	– WT-CA-TCN achieves the best performance when prices, blockchain information, social and interest data are considered.
Jakubik et al. (2023)	– Target: Bitcoin – Predictors: 980 Financial Times news articles with “bitcoin,”, “btc”, “Bitcoin or “BTC,” or “btc”	– Sentiment Labelled Sentences Data Set classifier – LSTM – CNN – Random Forest	– Machine learning models with financial news show the best performance. – Trading strategy using the proposed forecasting model is more profitable than a buy-and-hold strategy.
Liu et al. (2023)	– Target: 3703 cryptocurrencies – Predictors: Blockchain information, Google trends, Effective Federal Funds Rate, Unemployment Rate, Inflation, VIX, Gold, Market State Dynamic Commodity Index, PIMCO investment-grade bond ETF, S&P500, China Hushen300, Korea KOSPI, Crude oil, U.S. dollar index, EUR, Net generation of electricity of all sectors in the U.S, Electricity generation in China, Average retail electricity price in the U.S.	– OLS and XGB	– The predictive power of the 1-day lagged return is stronger than all other predictors combined. – Macroeconomic variables are important for forecasting cryptocurrency returns. – Stocks, commodities, and bonds are less effective for forecasting cryptocurrency returns. – Google searches are most effective for forecasting returns for larger cryptocurrencies. – The cost of mining has weak predictive effect.
Liu and Tsyvinski (2021)	– Target: Bitcoin, Ethereum, Ripple. – Predictors: network factors (number of wallet users, active addresses, transaction count, and Payment count), and production factors (U.S. electricity price growth, net U.S. generation growth, U.S. electricity consumption growth,	– Vector Autoregression (VAR)	– Cryptocurrency returns are influenced by network factors. – Cryptocurrency returns are not exposed to production factors. – Momentum and investor attention strongly predict cryptocurrency returns.

(*continued on next page*)

Table 1 (*continued*)

Authors	Variables	Models	Results
	the China generation growth, and Sichuan generation growth)		
Wang et al. (2023)	– Target: Bitcoin, Ethereum, Litecoin and Ripple volatilities – Predictors: Internal: lagged volatility, moving average volatility, open, high, low close prices, trading volume, trading count, and time, Blockchain, Google trend, Cryptocurrencies Fear & Greed Index. External: Adjusted close and volume of Gold, Oil, Silver, S&P 500, DJI, Russell 2000, NASDAQ, JPY-USD, USD-EUR and Policy uncertainty index.	– Random Forest – LSTM – Genetic Algorithm – Artificial Bee Colony – SHapley Additive exPlanations	– Random Forest and LSTM optimized by Genetic Algorithm and Artificial Bee Colony significantly outperform GARCH model. – Internal predictors are more important in volatility forecasting. – Predictors from several cryptocurrencies outperform those from a single cryptocurrency.
Sun et al. (2020)	– Target: 42 kinds of primary cryptocurrencies – Predictors: different stock markets, WTI crude oil	– LGBM	– LGBM outperforms SVM and Random Forest
Zhong et al. (2023)	– Target: Bitcoin – Predictors: cryptocurrency interrelations and trading volume, market capitalization, technical indicators, stock markets, commodities, fiat currencies, and Google trends	– LSTM and ReGAT	– LSTM-ReGAT provides better prediction performance compared with the linear regression, SVM, ANN, LSTM, and CNN models.

August 1, 2011, to July 31, 2018. They find that Bitcoin prices are influenced by blockchain information, public attention, stock prices, commodity prices, and fiat currency prices. Guo et al. (2021) use a Wavelet Transform (WT)-Casual multi-head Attention (CA)-Temporal Convolutional Network (TCN) and find the best forecasting performance when considering historical prices of Bitcoin, inter-exchange transaction data, and Google Trends from January 1, 2016, to December 31, 2018. Wang et al. (2022) adopt the LSTM approach to assess how investor attention impacts Bitcoin returns by constructing a comprehensive proxy for investor attention combining Google trends and tweets, as well as Bitcoin trading variables, and find that including these variables improves the prediction accuracy of Bitcoin returns. Liu et al. (2023) use the XGB model to forecast cryptocurrency returns using a range of different predictors, including cryptocurrencies and blockchain information variables, equities, commodities, bonds, currencies, macroeconomic indicators (unemployment rate, industrial production, and inflation), and Google search data for the period between April 28, 2013, and November 30, 2021. They find that the predictive power of the 1-day lagged return is stronger than that of all predictors combined. Zhong et al. (2023) propose a model combining LSTM and the Relation wise Graph Attention Network (ReGAT) to predict cryptocurrency price trends while incorporating cryptocurrency interrelations and trading volume, market capitalization, technical indicators, stock markets, commodities, fiat currencies, and Google Trends Index from March 30, 2020, to December 20, 2022. They find that the prediction performance was improved using LSTM-ReGAT compared with the linear regression, SVM, ANN, LSTM, and CNN models. Jakubik et al. (2023) focus on predicting Bitcoin returns using 980 Financial Times news articles and LSTM, CNN, and Random Forest models. They find that incorporating financial news significantly enhances the model performance during the period between September 12, 2013, and June 18, 2019, and that a trading strategy based on their proposed forecasting model is more profitable than a simple buy-and-hold strategy. Wang et al. (2023) study the volatilities predictability of Bitcoin, Ethereum, Litecoin, and Ripple using Random Forest and optimized LSTM models and find that the proposed LSTM model outperforms traditional GARCH models during the period spanning from November 1, 2017, to July 31, 2022. Cheng et al. (2024) highlight the superior performance of LSTM model, compared to Seasonal Autoregressive Integrated Moving Average (SARIMA) and Facebook Prophet models, in predicting Bitcoin prices over the period spanning from January 1, 2017, to October 30, 2022. Feng et al. (2024) employ a range of economic indicators, blockchain information, indices, commodities, and Google trends from January 1, 2017, to December 31, 2022, and show that the dynamic tuned regularization models (Lasso-tune, Ridge-tune, Ela-tune) outperform the Heterogeneous Autoregressive-Realized Volatility (HAR-RV) model, with hash rate and effective federal funds rate as the best predictors.

The above literature review reveals that previous studies ignore the impact of environmental variables on cryptocurrency return forecasting using ML models, do not consider the recent categorization of cryptocurrencies into green and dirty while identifying the relevant predictors, and most previous works based on ML use manual tuning and fixed parameters. Considering the predictive power of the DL and wavelet models, this study introduces a finetuned hybrid DWT-LSTM model to explore the role of each predictor among cryptocurrencies, commodities, blockchain information, Twitter-based economic uncertainty, and CO2 emissions in improving the forecasting accuracy of Bitcoin and Ripple returns.

3. Methodology

We introduce a hybrid forecasting model that optimizes the hyperparameters of the Long Short-Term Memory (LSTM) for each level of wavelet decomposition of the Maximum Overlapping Discrete Wavelet Transform (MODWT). Our model contributes to the previous work of Parvini et al. (2022) by finetuning the hyperparameters of the LSTM model for each level of wavelet decomposition of the

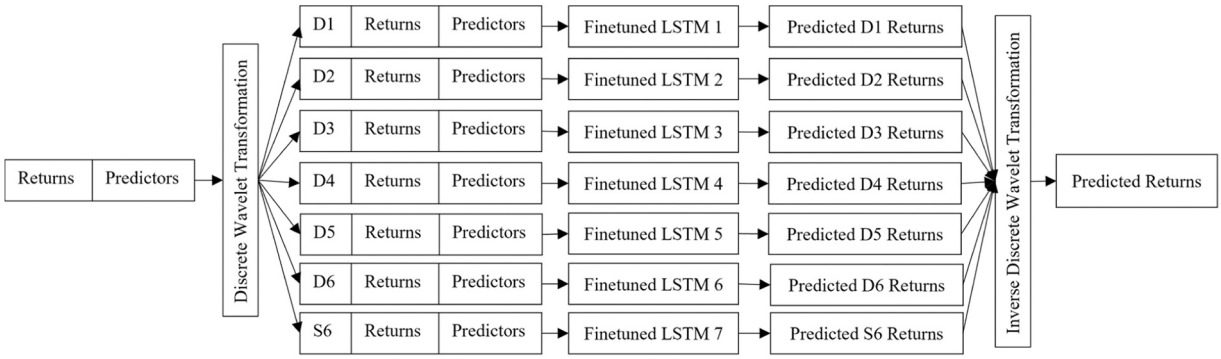

Fig. 1. : The framework of the proposed model.

DWT. This finetuning leads to more accurate and reliable predictions by improving generalization to new data, reducing overfitting, and ensuring efficient use of computational resources (Feng et al., 2024; Yang and Shami, 2020). We apply random search cross-validation of Bergstra and Bengio (2012) to find the best values of LSTM units,[2] fully connected layers,[3] neurons,[4] and dropout probability.[5] The forecasting rounds for each decomposition were performed using LSTM, then inverse MODWT is applied to generate the predicted returns. Fig. 1 illustrates the various steps of the forecasting model.

According to Fig. 1, the forecasting process is performed in three steps. First, the original data set, consisting of the desired cryptocurrency and predictors, is DWT decomposed up to six levels from D1 to D6, while S6 is the smooth subseries. Second, the LSTM model is finetuned, trained and tested on each decomposition as well as the smooth subseries (D1 to D6, and S6). Third, the inverse DWT is applied on the predicted D1 to D6, and S6 to generate the final predicted returns.

3.1. Long short-term memory (LSTM)

The LSTM architecture (Hochreiter and Schmidhuber, 1997) is a Recurrent Neural Network (RNN) engineered to avoid the issue of long-term dependencies. It is equipped with distinctive units known as memory blocks. Additionally, LSTM comprises several units referred to as "gates", which regulate the transmission of information.

The output of the blocks h_t is calculated using a filtered function of the current state of cell C_t and o_t. The output $Y = \{h_t\}$ is generated from an input $X = \{x_t\}$ for $t = 1, 2, \ldots, T$ employing the subsequent equations:

$$\begin{aligned} f_t &= \sigma\left(W^f[h_{t-1}, x_t] + b^f\right) \\ i_t &= \sigma\left(W^i[h_{t-1}, x_t] + b^i\right) \\ \widetilde{C}_t &= \tanh\left(W^c[h_{t-1}, x_t] + b^c\right) \\ C_t &= f_t \otimes C_{t-1} + i_t \otimes \widetilde{C}_t \\ o_t &= \sigma\left(W^o[h_{t-1}, x_t] + b^o\right) \\ h_t &= o_t \otimes \tanh(C_t) \end{aligned} \tag{1}$$

where $\otimes$ symbol represents an element-wise, $\boldsymbol{C_t}$ and $\boldsymbol{o_t}$ are the cell states, $\boldsymbol{i_t}$ and $\widetilde{\boldsymbol{C}}_{\boldsymbol{t}}$ represent potential entries for storage, $\boldsymbol{W}$ denotes the weights, $\boldsymbol{b}$ specifies the biases, $\boldsymbol{\sigma}$ and **tanh** stand for the sigmoid and hyperbolic tangent activation functions respectively. We use Adam Optimizer on mean square error (MSE) as the criterion for the loss.

3.2. Discrete wavelet transformation

Wavelet transformation techniques are frequently used to improve the accuracy of time series forecasting (Ortega and Khashanah, 2014; Barunik et al., 2016; Rua, 2017; Risse, 2019; Jana et al., 2021). Let $h_l = \{h_0, h_1, \ldots, h_{L-1}\}$ represents the wavelet filter with length L, h_l has the following characteristics:

[2] LSTM units are artificial neurons that remember values over time intervals.

[3] Fully connected layers are a fundamental component in neural networks where each neuron receives input from all neurons in the previous layer.

[4] A neuron takes a set of inputs, applies a transformation function, and produces an output.

[5] Dropout probability refers to the rate at which an artificial neuron is temporarily removed from the network during the training phase.

Table 2
Summary statistics of the selected data.

	Mean	Std	Min	Max	Skewness	Kurtosis
Cryptocurrencies						
Bitcoin	0.10 %	3.78 %	−23.80 %	18.43 %	−0.3410	4.7847
Ethereum	0.12 %	5.04 %	−31.48 %	31.05 %	−0.1905	4.9748
Ripple	−0.01 %	6.86 %	−50.86 %	60.75 %	0.5265	15.752
Stellar	−0.10 %	5.50 %	−24.14 %	45.56 %	0.9189	9.2938
Commodity indices						
S&P GSCI All Metals	0.03 %	0.95 %	−3.86 %	4.56 %	−0.1182	1.6851
S&P GSCI Agriculture	0.04 %	1.20 %	−5.25 %	4.84 %	−0.1310	1.9785
S&P GSCI Energy	0.02 %	2.79 %	−30.18 %	15.99 %	−1.7179	20.457
Bitcoin blockchain information						
Bitcoin block size	0.00 %	9.60 %	−38.47 %	39.33 %	−0.0871	1.7824
Bitcoin hashrate	0.15 %	8.89 %	−45.33 %	33.54 %	−0.1957	0.9848
Bitcoin transaction fees	0.06 %	19.04 %	−76.73 %	85.06 %	0.2130	1.7290
Bitcoin number of transactions	0.02 %	8.38 %	−49.23 %	38.98 %	0.1904	2.6278
Ripple blockchain information						
Ripple transaction fees	−0.12 %	45.67 %	−504.31 %	494.56 %	0.1732	32.043
Ripple number of transactions	0.07 %	13.01 %	−73.30 %	109.88 %	0.6102	7.7772
CO2 emissions	0.00 %	0.02 %	−0.09 %	0.06 %	−0.9047	4.1175
Twitter economic uncertainty (TEU-ENG)	0.13 %	23.47 %	−142.68 %	150.97 %	0.2958	4.1904

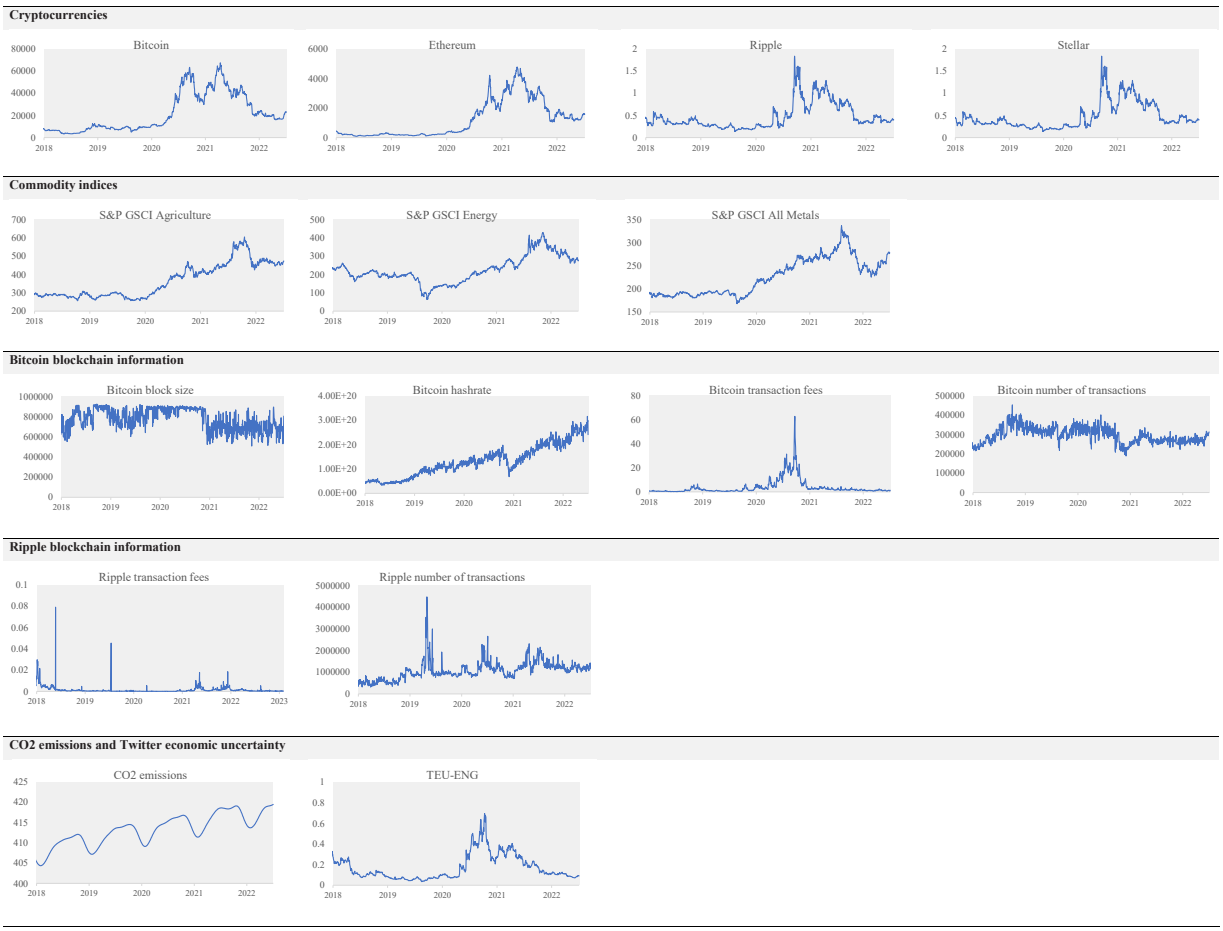

Fig. 2. : Evolution of the selected data during the period from August 01, 2018, to January 31, 2023.

Table 3
Hyperparameters of the proposed and benchmark models.

Models	Hyperparameters
RF	max_depth, max_features, min_samples_leaf, min_samples_split, n_estimators
XGB	max_depth, n_estimators, colsample_bytree, learning_rate, subsample
LGBM	colsample_bytree, learning_rate, max_depth, min_child_samples, n_estimators, subsample
ANN	Fully connected layers, Fully connected neurons, Activations, Dropout probability
CNN	Filters, Fully connected layers, Fully connected neurons, Dropout probability
LSTM	LSTM units, Fully connected layers, Fully connected neurons, Dropout probability

$$\sum_{l=0}^{1} h_l = 0, \sum_{l=0}^{1} h_l^2 = \frac{1}{2}$$
$$\sum_{l=-\infty}^{+\infty} h_l h_{l+2n} = 0, \quad n \neq 0 \tag{2}$$

The wavelet scaling filter g_l serves as a complement to the wavelet filter h_l, and has the following properties:

$$\sum_{l=0}^{1} g_l = 0, \quad \sum_{l=0}^{1} g_l^2 = \frac{1}{2}, \sum_{l=-\infty}^{+\infty} g_l g_{l+2n} = 0, \quad n \neq 0$$
$$\sum_{l=-\infty}^{+\infty} g_l h_{l+2n} = 0, \quad n \neq 0 \tag{3}$$

The wavelet and scaling filters are related to each other as following:

$$g_l = (-1)^{l+1} h_{L-1+l} \quad l = 0, \ldots, L-1 \tag{4}$$

The MODWT wavelet coefficients $\widetilde{W}_{j,t}$ and $\widetilde{V}_{j,t}$ at level j are defined as the convolution of the time series $X = \{x_t, t = 1, \ldots, N-1\}$ and are formulated as follows:

$$\widetilde{W}_{j,t} = \sum_{l=0}^{L_j-1} \widetilde{h}_{j,l} X_{t-l \bmod N}, \quad \widetilde{V}_{j,t} = \sum_{l=0}^{L_j-1} \widetilde{g}_{j,l} X_{t-l \bmod N} \tag{5}$$

where $\widetilde{\boldsymbol{h}}_{j,l} = \boldsymbol{h}_{j,l} \boldsymbol{2}^{-j/2}$ and $\widetilde{\boldsymbol{g}}_{j,l} = \boldsymbol{g}_{j,l} \boldsymbol{2}^{-j/2}$ are adjusted to length $\boldsymbol{N}$, $\boldsymbol{L_j} = (\boldsymbol{2j} - \boldsymbol{1})(\boldsymbol{L} - \boldsymbol{1}) + \boldsymbol{1}$.
The MODWT wavelet coefficients at each level can be represented as follows:

$$\widetilde{W}_j = \widetilde{w}_j X, \quad \widetilde{V}_j = \widetilde{v}_j X \tag{6}$$

Table 4
Evaluation metrics of the different models[14]

Model	Bitcoin			Ripple		
	MSE	MAE	R^2_{OS}(%)	MSE	MAE	R^2_{OS}(%)
Non-decomposed						
ANN	0.0023	0.0764	−43.27 %	0.0079	0.2624	−4.32 %
LSTM	0.0017	0.0565	−4.75 %	0.0054	0.1794	−30.74 %
CNN	0.0018	0.0598	−11.22 %	0.0063	0.2093	−35.70 %
XGB	0.0026	0.0864	−61.10 %	0.0067	0.2226	−2.63 %
RF	0.0016	0.0532	0.63 %	0.0036	0.1196	−45.31 %
LGBM	0.0023	0.0764	−42.22 %	0.0065	0.2159	−6.25 %
Decomposed						
DWT-ANN	0.0003	0.0100	70.33 %	0.0004	0.0133	85.91 %
DWT-LSTM	**0.0003**	**0.0100**	**75.17 %**	**0.0004**	**0.0133**	**85.97 %**
DWT-CNN	0.0009	0.0299	46.33 %	0.0018	0.0598	39.51 %
DWT-XGB	0.0004	0.0133	60.30 %	0.0007	0.0233	75.66 %
DWT-RF	0.0005	0.0166	56.20 %	0.0011	0.0365	63.79 %
DWT-LGBM	0.0004	0.0133	60.94 %	0.0009	0.0299	71.19 %

Note: Training set: August 01, 2018 - June 21, 2022, Validation set: June 22, 2022 - January 31, 2023. MSE, MAE and R^2_{OS} stand for Mean Square Error, Mean Absolute Error and the Out-of-Sample Coefficient of Determination, respectively

[14] All the selected variables are used as predictors.

Table 5
Evaluation metrics of the predictors of Bitcoin and Ripple returns.

Predictor	Bitcoin			Ripple		
	MSE	MAE	$R^2_{OS}(\%)$	MSE	MAE	$R^2_{OS}(\%)$
Cryptocurrencies						
Ethereum	0.00026	0.0135	76.49 %	0.000616	0.0320	84.31 %
Ripple (Bitcoin)	0.00027	0.0140	75.95 %	0.000706	0.0367	83.71 %
Stellar	0.00027	0.0140	75.62 %	0.000688	0.0358	84.09 %
Commodities						
S&P GSCI Agriculture	0.00028	0.0146	75.31 %	0.000644	0.0335	85.21 %
S&P GSCI Energy	0.00027	0.0140	75.84 %	0.000643	0.0334	85.38 %
S&P GSCI Metals	0.00028	0.0146	75.45 %	0.000705	0.0367	83.97 %
External						
Blockchain information	0.00024	0.0125	77.74 %	0.000653	0.0340	83.58 %
Twitter Economic Uncertainty	0.00021	0.0109	80.17 %	0.000651	0.0339	83.96 %
CO2 emissions	0.00022	0.0114	79.45 %	0.000665	0.0346	83.32 %

Note: Training set: August 01, 2018 - June 21, 2022, Validation set: June 22, 2022 - January 31, 2023. MSE, MAE and R^2_{OS} stand for Mean Square Error, Mean Absolute Error and the Out-of-Sample Coefficient of Determination, respectively

where:

$$\widetilde{w}_j = \frac{1}{2^j}\begin{pmatrix} \widetilde{h}_{j,0} & \widetilde{h}_{j,N-1} & \widetilde{h}_{j,N-2} & \dots & \widetilde{h}_{j,1} \\ \widetilde{h}_{j,1} & \widetilde{h}_{j,0} & \widetilde{h}_{j,N-1} & \dots & \widetilde{h}_{j,2} \\ \widetilde{h}_{j,2} & \widetilde{h}_{j,1} & \widetilde{h}_{j,0} & \dots & \widetilde{h}_{j,3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \widetilde{h}_{j,N-1} & \widetilde{h}_{j,N-2} & \widetilde{h}_{j,N-3} & \dots & \widetilde{h}_{j,0} \end{pmatrix}$$

and

$$\widetilde{v}_j = \frac{1}{2^j}\begin{pmatrix} \widetilde{g}_{j,0} & \widetilde{g}_{j,N-1} & \widetilde{g}_{j,N-2} & \cdots & \widetilde{g}_{j,1} \\ \widetilde{g}_{j,1} & \widetilde{g}_{j,0} & \widetilde{g}_{j,N-1} & \cdots & \widetilde{g}_{j,2} \\ \widetilde{g}_{j,2} & \widetilde{g}_{j,1} & \widetilde{g}_{j,0} & \cdots & \widetilde{g}_{j,3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \widetilde{g}_{j,N-1} & \widetilde{g}_{j,N-2} & \widetilde{g}_{j,N-3} & \cdots & \widetilde{g}_{j,0} \end{pmatrix}$$

The resulting quantities are:

$$\widetilde{D}_j = \widetilde{w}_j^T \widetilde{W}_j\,, \quad \widetilde{S}_j = \widetilde{v}_j^T \widetilde{V}_j \tag{7}$$

The inverse of MODWT decomposition is:

$$X = \sum_{j=0}^{J} \widetilde{w}_j^T \widetilde{W}_j + \widetilde{v}_j^T \widetilde{V}_j = \sum_{j=0}^{J} \widetilde{D}_j + \widetilde{S}_j \tag{8}$$

3.3. Random search for hyper-parameter optimization

Hyper-parameter optimization by random search is an efficient method to identify the optimal set of hyper-parameters for ML and DL models (Bergstra and Bengio, 2012). Unlike grid search (Burges, 1998) which systematically explores the parameter space on a fixed grid, random search randomly samples parameter values from predefined distributions, allowing to potentially cover a broader and more diverse range of the parameter space. This approach is particularly advantageous in high-dimensional spaces where exhaustive grid search becomes computationally impractical.

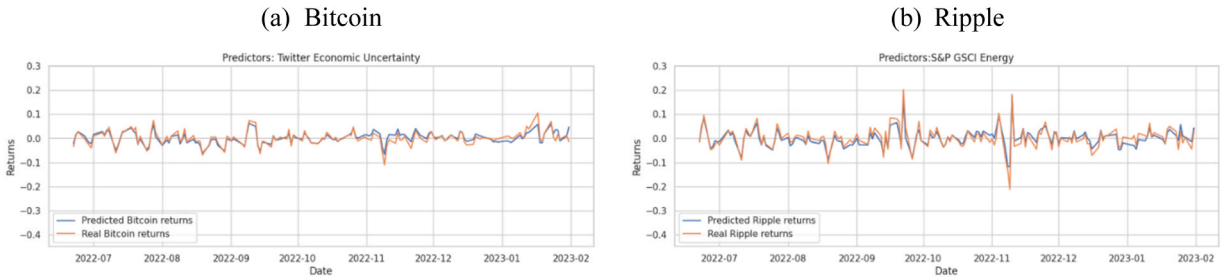

Fig. 3. : Actual vs predicted returns of Bitcoin and Ripple using the best predictors.

Table 6
Economic profitability of the trading strategy.

	Bitcoin		Ripple	
	Buy & hold	Forecasting based strategy	Buy & hold	Forecasting based strategy
Total Return	9 %	348 %	21 %	496 %
Max Drawdown	−2133 %	−8.54 %	−141 %	−16.78 %
Std	2.84 %	5.64 %	4.65 %	11.74 %
Sharpe Ratio	3.14	61.67	4.58	42.23
Sortino Ratio	4.55	231.29	6.49	59.79

4. Data and empirical results

Following the classification of cryptocurrencies into green and dirty cryptocurrencies (Ren and Lucey, 2022),[6] we forecast Bitcoin and Ripple returns considering a broad set of predictors. Our considered list of predictors includes other cryptocurrencies to consider the systematic risk,[7] commodity indices (S&P GSCI Energy, S&P GSCI All Metals, and S&P GSCI Agriculture) as proxies for the commodities markets, the Twitter-based Uncertainty Index (TEU-ENG)[8] as a proxy for economic factors, blockchain information for each cryptocurrency to measure the importance of technology factors in the forecasting, and CO2 emissions[9] to consider the impact of environmental factors. We selected daily data[10] covering August 01, 2018, to January 31, 2023. Table 2 summarizes the descriptive statistics of the selected variables.[11]

4.1. Forecasting results

The forecasting process consists of predicting one-day-ahead of Bitcoin and Ripple returns using a hybrid DWT-LSTM model developed using Tensorflow[12] on Python carried on Google Colab Pro.[13] We allocated 85 % of the observations in each subsample for the training and the remaining 15 % for the test. We opted for the Daubechies db3 filter (Daubechies, 1992) and set the decomposition level to six, based on the works of In and Kim (2012) and Parvini et al. (2022). In every forecasting cycle, the model employs five observations to predict one-day-ahead values. It then shifts one observation forward and repeats this process to forecast the subsequent observation. We compared the performance of the proposed model with those of the RF, XGB, LGBM, ANN, and CNN on wavelet-decomposed and non-decomposed data. We use random-search cross-validation to fine-tune the models hyperparameters. Table 3 summarizes the hyperparameters of the LSTM and other benchmark models. The search space is defined as a large grid in which each point corresponds to a specific combination of hyperparameter values. The combinations of these parameters were randomly searched to determine the optimal set of values that lead to the best performance on the validation datasets. The optimized parameters are presented in Tables A.1, A.2, A.3, and A.4 in the Appendix.

Table 4 shows the evaluation metrics (MSE, MAE, R^2_{OS}) of the proposed LSTM model and other benchmark models with both wavelet-decomposed and non-decomposed data for forecasting Bitcoin and Ripple returns. These metrics underline the performance

[6] Ren and Lucey (2022) argue that "proof of work" and "proof of stake" algorithms are responsible for high-and low-energy usage in mining these cryptocurrencies.
[7] As stated by Aslanidis et al., (2019); Ballis and Drakos, (2020); Bouri et al., (2020); Katsiampa et al., (2019a).
[8] TEU-ENG consists of the total number of daily English-language tweets containing both uncertainty terms as well as economy terms.
[9] The accumulation of carbon dioxide (CO2) in the atmosphere resulting from the combustion of coal, oil, and natural gas.
[10] Data are collected from https://bitinfocharts.com/, https://spglobal.com/, https://policyuncertainty.com/ and https://gml.noaa.gov/ccgg/trends/.
[11] Log returns are considered for the financial markets.
[12] https://www.tensorflow.org/
[13] https://colab.google/

enhancement when wavelet decomposition is applied and show that our proposed hybrid DWT-LSTM model outperforms all other benchmark models.

In fact, the additional layer of DWT to LSTM model shows the ability to capture multiscale features by effectively capturing both short-term fluctuations and long-term trends while reducing noise and redundancy. These findings align closely with the results of Wang et al. (2022), and Cheng et al. (2024) on the performance of LSTM model as time series forecasting model, and the conclusions of Parvini et al. (2022), affirming the value of integrating DWT with LSTM for robust feature extraction in time series.

Table 5 summarizes the impact of the considered predictors (cryptocurrencies, commodities, blockchain information, Twitter economic uncertainty, and CO2 emissions) on the forecasting performance[15] of clean and dirty cryptocurrency returns. Our results underline the comparable performance of the considered predictors, with the Twitter Economic Uncertainty index being the best predictor of Bitcoin returns and S&P GSCI Energy being the best predictor of Ripple returns. The predictive power of other cryptocurrencies can be explained by the strong links between cryptocurrencies and their herding behavior (Chen et al., 2024; Bouri et al., 2022; Aslanidis et al., 2019; 2020). The predictive power of commodities can be explained by the links between cryptocurrencies and commodities (Nakagawa and Sakemoto, 2022; Sarkodie et al., 2022; Ji et al., 2019; Hayes, 2017; Bouri et al., 2017b; Jebabli et al., 2014; Baur et al., 2018; Dyhrberg, 2016a, 2016b; Qiu et al., 2021; Naeem et al., 2023). The importance of technological factors in forecasting Bitcoin and Ripple returns reinforces the findings of Balcilar et al. (2017), Guo et al. (2021), Chen et al. (2021) and Feng et al. (2024). In fact, blockchain information variables are important in cryptocurrencies pricing. A higher hash rate indicates increased miner investment and network security, whereas a larger block size enhances transaction efficiency, and the transaction volume provides information about cryptocurrencies liquidity (Balcilar et al., 2017; Guo et al., 2021). The forecasting capability of Twitter Economic Uncertainty reinforces the findings of Kraaijeveld and de Smedt (2020), Dias et al. (2022), Hajek et al. (2023), Wu et al. (2021), and Aharon et al. (2022) and can be explained by the rise in popularity of social media platforms, particularly the Twitter-based economic uncertainty. The predictive power of CO2 emissions underlines the role of environmental factors, mainly for Ripple returns forecasting, which can be explained by the lower energy consumption of Ripple compared to Bitcoin (Ren and Lucey, 2022; Pham et al., 2022; Zhang et al., 2023; Clark et al., 2023). This drives investors to switch to clean cryptocurrencies when carbon emissions are high, as investors rush to support sustainable and eco-friendly options in response to environmental concerns (Duan et al., 2023; Sharif et al., 2023) as part of the ESG[16] investment transition.

Fig. 3 shows the actual versus predicted returns using the best predictors for Bitcoin and Ripple. The predicted and real returns align perfectly, which means that our model and the selected predictors are dynamically accurate.

4.2. Robustness check

To check the robustness of our forecasting model, we re-evaluated the model using a different sample split (70 % for training, 30 % for testing). Despite the change in the proportion of training and testing data, the forecasting results remained unchanged (see Tables A.6, A.7 and A.8 in the Appendix). This consistency across different sample splits indicates that DWT-LSTM is robust, reliable, and insensitive to data partition.

4.3. Implications on trading

To examine the implications of our returns forecasting on cryptocurrencies trading performance, we perform a trading strategy based on the forecasted returns (the trader initiates "Buy" signal at time t if the forecasted return for time $t+1$ is positive and opts to "Sell" otherwise). We compare this strategy with the simple "buy-and-hold approach."[17] Table 6 highlights that our trading strategy for both Bitcoin and Ripple yielded remarkable results, substantially outperforming the buy-and-hold approach, with significantly higher total returns and superior risk-adjusted performance. Our proposed strategy allows for better risk management, as evident from the considerably lower maximum drawdowns and impressive Sharpe and Sortino ratios. Additionally, Figures A.1 and A.2 in the Appendix highlight that our proposed trading strategy allows to avoid price drops and provides much higher cumulative returns in simulated trading than the buy & hold strategy.

5. Conclusion

This study investigates several predictors of Bitcoin and Ripple returns (cryptocurrencies, commodities, blockchain information, Twitter eco-economic uncertainty, and CO2 emissions). We introduce a hybrid DWT-LSTM model to forecast one-day-ahead returns. Our results demonstrate that the proposed DWT-LSTM model surpasses the selected benchmark models (Random Forest (RF), XGBoost (XGB), LightGBM (LGBM), Artificial Neural Networks (ANN), and Convolutional Neural Networks (CNN)) in terms of forecasting accuracy. We also highlight the comparable performance of the considered predictors, with the Twitter Economic Uncertainty index being the best predictor of Bitcoin returns, while S&P GSCI Energy is the best predictor of Ripple returns.

Our findings underline the implications of forecasting Bitcoin and Ripple returns on their trading performance compared to a

[15] Multivariate prediction combining several predictors is presented in Table A.5 in the Appendix.

[16] ESG stands for Environmental, Social, and Governance, and it refers to a set of criteria that investors use to evaluate their investments based on their environmental and social impact, as well as their corporate governance practices.

[17] "Buy and hold" is a passive strategy that produces a stable portfolio over an extended period to provide higher returns.

simple buy-and-hold approach. The forecasting-based strategy for Bitcoin and Ripple significantly outperformed, yielding higher total returns and better risk-adjusted performance, lower maximum drawdowns, and higher Sharpe and Sortino ratios, which underlines its usefulness as a trading support system. In the other hand, our results hold paramount importance for policymakers as they offer invaluable insights into cryptocurrency market trends. These insights aid in anticipating the potential economic and environmental factors that drive different types of cryptocurrencies to swiftly adapt policies in response to the dynamic nature of cryptocurrencies in order to enhance regulation and compliance, protect investors, develop monetary and taxation policies, foster innovation, promote international cooperation, and ensure a balanced and stable financial system.

Future studies should focus on enhancing the accuracy and interpretability of the DWT–LSTM model. Exploring advanced feature engineering, such as integrating market sentiment analysis and macroeconomic indicators, can refine predictions.

Data availability

Data will be made available on request.

Appendix

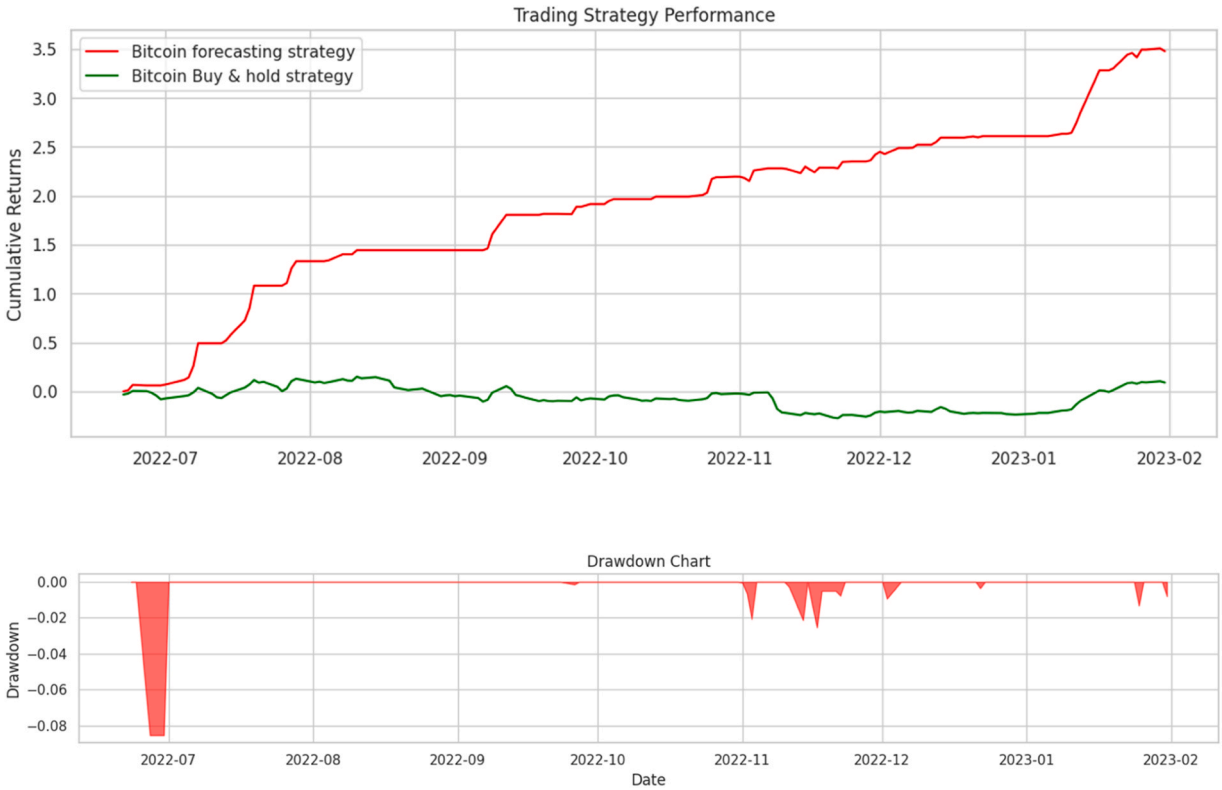

Figure A.1. Cumulative returns for the portfolio implementing a trading strategy based on the best predictor of Bitcoin (Twitter Economic Uncertainty Index).

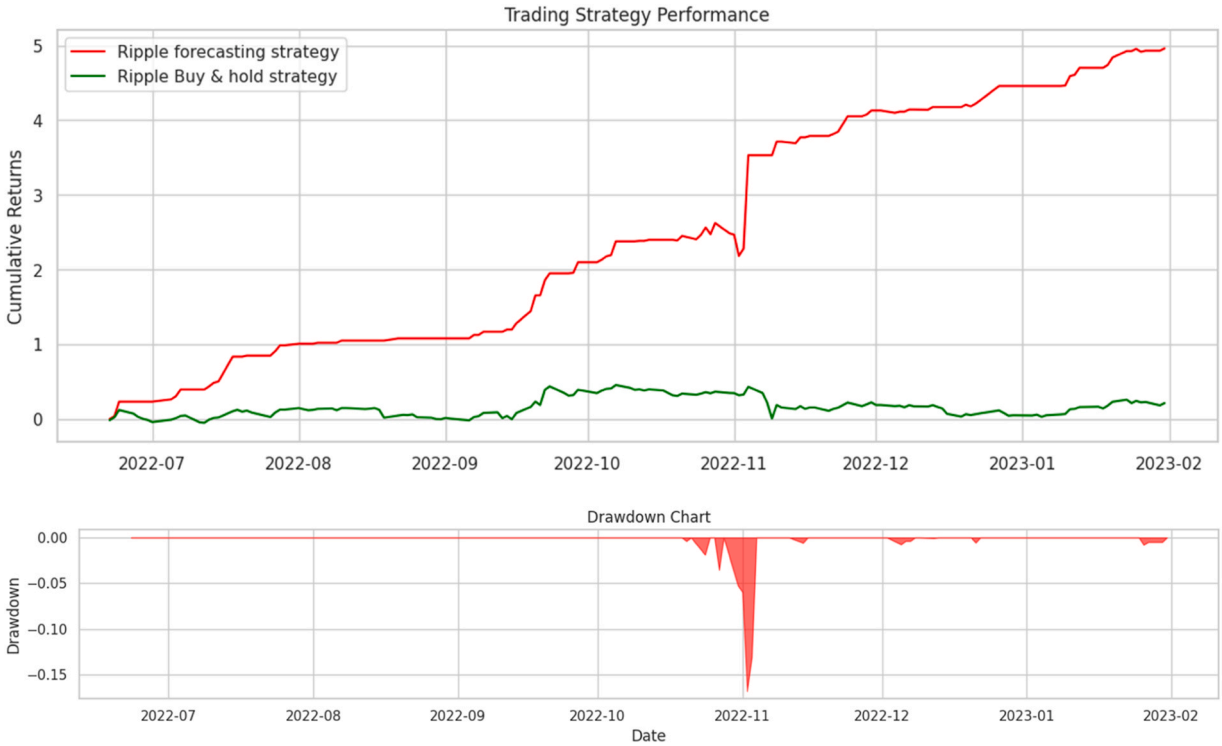

Figure A.2. Cumulative returns for the portfolio implementing a trading strategy based on the best predictor of Ripple (S&P GSCI Energy).

Table A.1
Optimized hyperparameters of non-decomposed models (RF, XGB, LGBM)

	Bitcoin			**Ripple**		
Hyperparameters	RF	XGB	LGBM	RF	XGB	LGBM
'max_depth'	10	None	10	10	None	10
'max_features'	'log2'	-	-	'log2'	-	-
'min_samples_leaf'	3	-	-	4	-	-
'min_samples_split'	4	-	-	9	-	-
'n_estimators'	185	449	120	825	763	324
'colsample_bytree'	-	0.543	0.733	-	0.611	0.751
'learning_rate'	-	0.015	0.015	-	0.022	0.015
'subsample'	-	0.770	0.725	-	0.971	0.817
'min_child_samples'	-	-	11	-	-	36

Table A.2
Optimized hyperparameters of non-decomposed models (ANN, LSTM, CNN)

	Bitcoin	**Ripple**
ANN		
Fully connected layers	3	5
Fully connected neurons	32	64
Activations	None	linear
Dropout probability	0	0.5
LSTM		
LSTM units	128	64
Fully connected layers	2	2
Fully connected neurons	16	32
Dropout probability	0	0.2
CNN		
Filters	64	32
Fully connected layers	3	3
Fully connected neurons	32	16
Dropout probability	0	0.5

Table A.3
Optimized hyperparameters of decomposed models (RF, XGB, LGBM)

Hyperparameters	Bitcoin							Ripple						
	D1	D2	D3	D4	D5	D6	S1	D1	D2	D3	D4	D5	D6	S1
RF														
max_depth	30	20	10	20	20	30	10	10	30	30	None	10	30	None
max_features	auto	auto	auto	auto	auto	auto	auto	log2	auto	auto	auto	auto	auto	auto
min_samples_leaf	4	4	3	3	4	1	3	4	4	4	2	2	4	2
min_samples_split	2	2	2	8	2	9	2	9	6	5	9	5	5	6
n_estimators	271	660	101	963	660	260	10	825	195	521	824	127	521	915
XGB														
max_depth	None	None	None	None	20	20	10	None	None	None	None	None	30	20
n_estimators	449	882	376	376	906	906	661	449	953	624	376	644	732	413
colsample_bytree	0.543	0.700	0.916	0.916	0.986	0.986	0.917	0.543	0.785	0.710	0.916	0.788	0.953	0.578
learning_rate	0.015	0.015	0.031	0.031	0.106	0.106	0.027	0.015	0.020	0.036	0.031	0.049	0.037	0.108
subsample	0.770	0.728	0.809	0.809	0.749	0.749	0.831	0.770	0.679	0.541	0.809	0.729	0.909	0.930
LGBM														
colsample_bytree	0.944	0.751	0.789	0.944	0.974	0.965	0.977	0.751	0.809	0.540	0.878	0.789	0.944	0.693
learning_rate	0.045	0.015	0.014	0.045	0.107	0.096	0.101	0.015	0.020	0.053	0.033	0.014	0.045	0.095
max_depth	20	10	10	20	20	10	30	10.000	30.000	None	None	10.000	20.000	None
min_child_samples	16	36	26	16	13	17	45	36	17	5	11	26	16	7
n_estimators	108	324	413	108	445	378	136	324	267	157	798	413	108	227
subsample	0.881	0.817	0.930	0.881	0.549	0.877	0.964	0.817	0.536	0.958	0.847	0.930	0.881	0.686

Table A.4
Optimized hyperparameters of decomposed models (ANN, LSTM, CNN)

Hyperparameters	Bitcoin							Ripple						
	D1	D2	D3	D4	D5	D6	S1	D1	D2	D3	D4	D5	D6	S1
ANN														
Fully connected layers	3	3	1	3	1	1	1	3	1	1	1	2	3	1
Fully connected neurons	32	32	32	32	32	32	32	32	32	32	32	32	32	32
Activations	None	linear	linear	linear	None	linear	None	linear	linear	None	linear	linear	None	None
Dropout probability	0	0	0.1	0	0.2	0.1	0.3	0	0	0	0.3	0.1	0.1	0
LSTM														
LSTM units	128	128	128	128	128	128	32	64	32	128	128	64	128	128
Fully connected layers	2	3	1	1	1	1	3	1	1	1	3	2	3	1
Fully connected neurons	16	32	16	32	16	32	16	32	16	16	32	32	32	32
Dropout probability	0	0.1	0	0	0.1	0.2	0	0.3	0.3	0	0	0.2	0.2	0.3
CNN														
Filters	64	32	32	64	64	32	16	32	6	32	32	32	32	16
Fully connected layers	3	3	1	3	1	1	1	2	3	1	2	1	1	1
Fully connected neurons	32	32	32	32	32	32	32	16	32	32	32	32	4	4
Dropout probability	0	0.1	0	0.2	0.1	0	0.2	0	0	0	0.3	0.1	0	0.1

Table A.5
Evaluation metrics of some combinations of predictors of Bitcoin and Ripple returns

Predictors	Bitcoin			Ripple		
	MSE	MAE	$R^2_{OS}(\%)$	MSE	MAE	$R^2_{OS}(\%)$
Combination of cryptocurrencies						
Ripple + Stellar (Bitcoin + Stellar)	0.000284	0.0148	74.63 %	0.000395	0.0205	84.80 %
Ethereum + Stellar	0.000350	0.0182	68.78 %	0.000395	0.0205	84.80 %
Ethereum + Ripple (Bitcoin + Ethereum)	0.000264	0.0137	76.47 %	0.000396	0.0206	85.06 %
Ethereum + Ripple + Stellar (Bitcoin + Ethereum + Stellar)	0.000282	0.0147	74.85 %	0.000391	0.0203	84.03 %
Combination of commodity indices						
Agriculture + Metals	0.000268	0.0139	76.06 %	0.000393	0.0204	84.48 %
Energy + Agriculture	0.000268	0.0139	76.11 %	0.000394	0.0205	84.67 %
Energy + Metals	0.000280	0.0146	75.06 %	0.000391	0.0203	84.10 %
Agriculture + Metals + Energy	0.000273	0.0142	75.65 %	0.000392	0.0204	84.26 %
Combination of all considered financial markets						
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy	0.000266	0.0138	76.32 %	0.000393	0.0204	84.48 %
Impact of the consideration of Blockchain information, Twitter Economic Uncertainty index, and CO2 emissions[a]						
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy + BI	0.000268	0.0139	76.10 %	0.000272	0.0141	84.22 %

(*continued on next page*)

Table A.5 (*continued*)

Predictors	Bitcoin			Ripple		
	MSE	MAE	$R^2_{OS}(\%)$	MSE	MAE	$R^2_{OS}(\%)$
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy + TEU	0.000258	0.0134	77.05 %	0.000290	0.0151	83.16 %
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy + CO2 emissions	0.000269	0.0140	76.06 %	0.000257	0.0134	85.06 %

[a]We consider the impact of adding each one of these variables as predictor to the combination of all selected financial markets.

Training set: August 01, 2018 - June 21, 2022, Validation set: June 22, 2022 - January 31, 2023. MSE, MAE and R^2_{OS} stand for Mean Square Error, Mean Absolute Error and the Out-of-Sample Coefficient of Determination, respectively.

Table A.6
Evaluation metrics of the different models (70 % training, 30 % validation)

Model	Bitcoin			Ripple		
	MSE	MAE	$R^2_{OS}(\%)$	MSE	MAE	$R^2_{OS}(\%)$
Non-decomposed						
ANN	0.0026	0.1193	−42.45 %	0.0128	0.3932	−3.97 %
LSTM	0.0019	0.0882	−4.66 %	0.0087	0.2687	−28.23 %
CNN	0.0020	0.0934	−11.01 %	0.0102	0.3135	−32.79 %
XGB	0.0029	0.1349	−59.95 %	0.0108	0.3334	−2.42 %
RF	0.0018	0.0830	0.62 %	0.0058	0.1792	−41.62 %
LGBM	0.0026	0.1193	−41.42 %	0.0105	0.3235	−5.74 %
Decomposed						
DWT-ANN	0.0003	0.0156	69.00 %	0.0007	0.0199	78.91 %
DWT-LSTM	**0.0003**	**0.0156**	**73.75 %**	**0.0007**	**0.0199**	**78.96 %**
DWT-CNN	0.0010	0.0467	45.45 %	0.0029	0.0896	36.29 %
DWT-XGB	0.0004	0.0208	59.16 %	0.0011	0.0348	69.49 %
DWT-RF	0.0006	0.0259	55.14 %	0.0018	0.0547	58.59 %
DWT-LGBM	0.0004	0.0208	59.79 %	0.0015	0.0448	65.39 %

Training set: August 01, 2018 - October 25, 2021, Validation set: October 26, 2021 - January 31, 2023. MSE, MAE and R^2_{OS} stand for Mean Square Error, Mean Absolute Error and the Out-of-Sample Coefficient of Determination, respectively.

Table A.7
Evaluation metrics of the predictors of Bitcoin and Ripple returns. (70 % training, 30 % validation)

Predictor	Bitcoin			Ripple		
	MSE	MAE	$R^2_{OS}(\%)$	MSE	MAE	$R^2_{OS}(\%)$
Cryptocurrencies						
Ethereum	0.00031	0.0141	73.35 %	0.00064	0.0194	76.43 %
Ripple (Bitcoin)	0.00034	0.0147	71.05 %	0.00071	0.0202	74.14 %
Stellar	0.00029	0.0137	75.08 %	0.00069	0.0200	74.80 %
Commodities						
S&P GSCI Agriculture	0.00032	0.0143	72.74 %	0.00064	0.0195	76.39 %
S&P GSCI Energy	0.00029	0.0137	75.04 %	0.00062	0.0190	77.44 %
S&P GSCI Metals	0.00030	0.0139	74.41 %	0.00070	0.0203	74.16 %
External						
Blockchain information	0.00031	0.0141	73.63 %	0.00065	0.0197	76.05 %
Twitter Economic Uncertainty	0.00029	0.0136	75.33 %	0.00065	0.0196	76.15 %
CO2 emissions	0.00031	0.0140	73.95 %	0.00067	0.0197	75.64 %

Note: Training set: August 01, 2018 - October 25, 2021, Validation set: October 26, 2021 - January 31, 2023. MSE, MAE and R^2_{OS} stand for Mean Square Error, Mean Absolute Error and the Out-of-Sample Coefficient of Determination, respectively.

Table A.8
Evaluation metrics of some combinations of predictors of Bitcoin and Ripple returns (70 % training, 30 % testing)

Predictors	Bitcoin			Ripple		
	MSE	MAE	$R^2_{OS}(\%)$	MSE	MAE	$R^2_{OS}(\%)$
Combination of cryptocurrencies						
Ripple + Stellar (Bitcoin + Stellar)	0.00032	0.0150	73.22 %	0.00040	0.0122	77.89 %
Ethereum + Stellar	0.00039	0.0185	67.48 %	0.00040	0.0122	77.89 %
Ethereum + Ripple (Bitcoin + Ethereum)	0.00029	0.0139	75.02 %	0.00040	0.0122	78.13 %

(*continued on next page*)

Table A.8 (*continued*)

Predictors	Bitcoin			Ripple		
	MSE	MAE	R^2_{OS}(%)	MSE	MAE	R^2_{OS}(%)
Ethereum + Ripple + Stellar (Bitcoin + Ethereum + Stellar)	0.00031	0.0149	73.43 %	0.00039	0.0121	77.18 %
Combination of commodity indices						
Agriculture + Metals	0.00030	0.0141	74.62 %	0.00040	0.0121	77.60 %
Energy + Agriculture	0.00030	0.0141	74.67 %	0.00040	0.0122	77.77 %
Energy + Metals	0.00031	0.0148	73.64 %	0.00039	0.0121	77.25 %
Agriculture + Metals + Energy	0.00030	0.0144	74.22 %	0.00039	0.0121	77.39 %
Combination of all considered financial markets						
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy	0.00030	0.0140	74.87 %	0.00040	0.0121	77.60 %
Impact of the consideration of Blockchain information, Twitter Economic Uncertainty index, and CO2 emissions[1]						
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy + BI	0.00030	0.0141	74.66 %	0.00027	0.0084	77.36 %
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy + TEU	0.00029	0.0136	75.59 %	0.00029	0.0090	76.38 %
Ethereum + Ripple + Stellar + Agriculture + Metals + Energy + CO2 emissions	0.00030	0.0142	74.62 %	0.00026	0.0080	78.13 %

Training set: August 01, 2018 - October 25, 2021, Validation set: October 26, 2021 - January 31, 2023. MSE, MAE and R^2_{OS} stand for Mean Square Error, Mean Absolute Error and the Out-of-Sample Coefficient of Determination, respectively.

[1]We consider the impact of adding each one of these variables as predictor to the combination of all selected financial markets.

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